

Comparative Analysis of Machine Learning Models for Heart Disease Prediction

Project Report

Abstract

This project aims to develop a reliable machine learning system to predict the presence of heart disease in patients. The study evaluates five different algorithms—Logistic Regression, Decision Tree, Support Vector Machine (SVM), Random Forest, and Gradient Boosting—to determine the most effective model for diagnosis. The methodology involves a hyperparameter tuning process using Grid Search, Random Search, and Bayesian Optimization, alongside a comparison of three categorical encoding strategies. Results indicate that the Support Vector Machine (SVM) outperformed other models, achieving the highest accuracy (90.16%) and a strong recall rate, making it the optimal choice for this medical diagnostic task.

1. Introduction

1.1 Project Structure

To ensure an organized and unbiased analysis, the project was structured with separate environments for each algorithm.

- **Organization:** A separate notebook was maintained for each algorithm to isolate results.
- **Visualization:** The final results represent the best performance achieved after testing three different encoding methods (One-Hot, Label, Target) and three hyperparameter search algorithms.

2. Data Preprocessing and Feature Engineering

2.1 Handling Categorical Data

processed using three distinct methods:

1. **One-Hot Encoding**
2. **Label Encoding**
3. **Target Encoding**

2.2 Feature Scaling

Medical features vary drastically in range Models can be biased toward larger numbers. To prevent features with larger magnitudes from dominating the learning process, scaling was applied.

- **Selected Method:** StandardScaler
- **Alternative Methods Considered:**
 - Min-MaxScaler
 - RobustScaler

3. Methodology

3.1 Algorithms and Models

The following classifiers were implemented and compared:

4. Logistic Regression
5. Decision Tree
6. Support Vector Machine (SVM)
7. Random Forest
8. Gradient Boost

3.2 Hyperparameter Optimization

9. **GridSearchCV:** Exhaustively searches through a manually specified subset of the hyperparameter space. Thorough for discrete values but computationally expensive.
10. **RandomizedSearchCV:** Samples a fixed number of parameter settings from specified distributions. Uses the least computational power but often yields less optimal results compared to exhaustive methods.
11. **Bayesian Optimization:** A probabilistic model that 'learns' from previous parameter evaluations to find the best values, including continuous values between grid points. Highly efficient and can discover parameter values that Grid Search would miss.

4. Results and Evaluation

4.1 Evaluation Metrics

The models were evaluated using the following metrics:

- **Accuracy:** The ratio of correctly predicted diagnoses.
- **Precision & Recall:** To understand the trade-off between false alarms (Precision) and missed diagnoses (Recall/False Negatives).
- **F1-Score:** The harmonic mean of Precision and Recall.
- **ROC-AUC:** The area under the Receiver Operating Characteristic curve, measuring the ability to distinguish between diseased and healthy patients.

4.2 Performance Comparison

Model Name	Accuracy	ROC-AUC	Recall (1)	Precision (1)	F1-Score (1)	Avg Precision
SVM	0.901639	0.956710	0.857143	0.923077	0.888889	0.961581
Random Forest	0.885246	0.946970	0.821429	0.920000	0.867925	0.944679
Logistic Regression	0.868852	0.957792	0.857143	0.857143	0.857143	0.959418
Random Forest (GB)	0.868852	0.941558	0.857143	0.857143	0.857143	0.933808
Decision Tree	0.803279	0.873377	0.785714	0.785714	0.785714	0.812151

4.3 Analysis

- **Best Performer:** The SVM model achieved the highest overall Accuracy (90.1%) and F1-Score (0.88). It demonstrated a superior balance of Precision and Recall compared to the other models.
- **Runner Up:** The Random Forest model followed closely with 88.5% accuracy.
- **Baseline:** The Decision Tree performed the poorest across all metrics (Accuracy ~80%), highlighting the value of ensemble methods (Random Forest) and complex margin-based classifiers (SVM) over simple trees.

5. Conclusion

The comparative analysis demonstrates that the Support Vector Machine (SVM) is the most effective model for this Heart Disease Prediction project. While Random Forest and Logistic Regression provided competitive results, SVM is the 'better model' for this specific medical dataset.